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Lecture 5: Vulnerabilities in Cyber Security

1. What is a Vulnerability?

A **vulnerability** is a **weakness, flaw, or gap** in a system, network, application, or process that can be **exploited by a threat actor** to compromise security.

Vulnerabilities may arise from:

- Design flaws
- Implementation errors
- Misconfigurations
- Human mistakes

A vulnerability by itself does not cause harm; **harm occurs when a threat exploits it**.

2. Common Areas of Vulnerability

2.1 Hardware Vulnerabilities

- **Outdated devices** are systems running old hardware or software that no longer receive security updates or vendor support.

- **Physical access to systems** means an unauthorized person can **touch, open, connect to, or control** a device directly.

- **Lack of hardware security controls** This means missing built-in hardware protections that defend systems against tampering, unauthorized access, and low-level attacks.

2.2 Software Vulnerabilities

- **Bugs in applications**
- **An unpatched operating system** is one that has not received or installed the latest security updates released by the vendor (e.g., Microsoft, Linux distributions, Apple).
- **Insecure code** means programming code that has weaknesses or flaws that attackers can exploit to break into a system, steal data, or cause damage.

2.3 Network Vulnerabilities

- Misconfigured firewalls
- Insecure protocols (FTP, Telnet)
- Poor network monitoring

2.4 Human Vulnerabilities

- Weak passwords
- Social engineering attacks
- Lack of security awareness

3. Impact Caused Due to Vulnerabilities

If vulnerabilities are exploited, the following impacts may occur:

- Data breaches and information leakage
- Financial loss
- Service disruption
- Loss of customer trust and reputation
- Legal and regulatory penalties

4. Risk in Cyber Security

In cybersecurity, **risk** refers to the probability that a threat will successfully exploit a system vulnerability.

Risk management involves:

- Identifying vulnerabilities
- Assessing threats
- Implementing controls
- Continuous monitoring

5. Types of Vulnerabilities: Vulnerabilities can be classified into:

a) Technical Vulnerabilities

- Software bugs
- Weak encryption
- Misconfigurations

b) Administrative Vulnerabilities

- Poor security policies
- Inadequate training

c) Physical Vulnerabilities

physical environment (buildings, rooms, equipment areas) is not properly protected, making it easier for unauthorized people to access systems or sensitive information.